

Pranav Achar

Dublin, CA | achar.pranav@gmail.com | github.com/PranavAchar01 | [linkedin.com/in/pranavachar](https://www.linkedin.com/in/pranavachar) | pranavachar.vercel.app

EDUCATION

Los Angeles Mission College, A.S. Computer Science & Mathematics (transfer track) **Expected transfer: Fall 2027**
4.0 GPA at LA Mission College. Also coursework at Diablo Valley College and Coastline College. 2026 to present
Coursework: Calculus I and II, Programming in C++; *in progress:* Calculus III, Advanced C/C++; *Spring 2027:* Linear Algebra & Differential Equations, Discrete Mathematics, Data Structures.

PROJECTS AND AWARDS

Player Two [\[code\]](#) [\[live\]](#) | **1st Place**, Track 1, The Executable World AI Hackathon Sep 2026

- Built an agent that turns a plain-language task into robot training data: it finds Creative Commons video, tracks pose with MediaPipe, and retargets it onto Unitree G1, SO-101 and Franka Panda robots via inverse kinematics in MuJoCo.
- One unattended run screened 68 videos and accepted 22 in under 17 minutes, exporting a LeRobot v3.0 dataset. Every episode must pass deterministic gates (85%+ pose tracking, joint limits, self-collision).
- In progress: driving a physical SO-101 robot arm from the same pipeline.

SixthSense [\[code\]](#) | **2nd Place**, ExecuTorch Hackathon (Qualcomm x Meta) Jun 2026

- On-device navigation copilot for blind users, [featured on the PyTorch Foundation blog](#). Built the ExecuTorch vision pipeline and LLM voice agent on a Galaxy S25 that runs Depth-Anything-V2, YOLOv11, Qwen2.5 and OCR fully offline.

Foundry [\[code\]](#) [\[live\]](#) | **1st Place**, Superserve Track, terac One-Person-Company Hackathon Aug 2026

- Built an agent that reads a real audience, writes a business for each person in it, then builds and deploys every storefront and runs outreach on a schedule. Spend ceilings, an append-only ledger and allowlisted outreach are enforced in code.

HealthFlow [\[code\]](#) | **1st Place**, Scalekit x Apify Hackathon May 2026

- Built a 9-agent system that turns a paramedic's voice capture into a committed EHR record in under 60 seconds, with differential diagnosis, drug-interaction safety blocks and a SHA-256 audit trail. 24 of 24 tests pass.

1st Place, Apify Track, Build Club Gemini x Exa Hackathon: trending X topics to 3D models (Hunyuan3D).

Finalist (top 6 of 115 teams), Abridge x Anthropic x Lightspeed Hackathon, with Vigil. Jul 2026

RESEARCH

Independent Researcher, markerless motion capture measurement study Sep 2026 to present

- Designing a preregistered study of how skip-frame person detection, a speed-up option, shifts joint angles in open-source markerless motion capture (rtmlib, RTMPose). In progress; preprint (not peer-reviewed) planned for November 2026.
- Traced the cause in rtmlib's code: between detections the person box is rebuilt from the last frame's keypoints, which snap to a 0.5-pixel grid, so the loop never settles; freezing the box makes the output identical. Pilot (10 still photos): skip-frame detection shifted joint angles 4.2° vs 1.1° from invisible image noise alone.

AI Research Intern, UC Santa Cruz AIEA Lab: multi-step logical reasoning in LLMs. May 2026

EXPERIENCE

Co-Founder, Optivia Mar 2025 to present

- Building a macOS app (Electron, React, TypeScript) that turns a plain-language idea into a dependency-aware task graph, staffs each node with a specialized agent, and runs the agents in parallel on LangGraph.
- Built the training-data flywheel for its computer-use agents: runs tasks in fresh VMs, scores each episode with programmatic verifiers, and exports winners as a fine-tuning (SFT) dataset.

Co-Founder and CEO, Web Design Initiative (nonprofit) 2024 to present

- Lead 6 engineers delivering pro-bono websites to under-resourced organizations; 8+ client organizations in the first year.

Engineering Intern, Opterra Energy Solutions Jun to Aug 2026

- Wrote a Heat-Recovery-as-a-Service proposal to sell AI data center waste heat as district heat, with the system design and a financial model (about \$3.5M/yr per 10 MW site, 2 to 4 year payback).

OPEN SOURCE

11 merged pull requests to [statsmodels](#) (8), [LanceDB](#) (1) and [Biome](#) (2): crash fixes and GLM docs, a Python-client fix, and Rust CSS-parser fixes.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Kotlin, Java, SQL, Rust

ML and robotics: PyTorch, ExecuTorch, pose estimation (MediaPipe, RTMPose), YOLO, MuJoCo, LeRobot, NumPy

Agents and systems: Anthropic SDK (Claude), LangGraph, LangChain, Next.js, React, Node.js, Postgres, Docker